

DAV Quiz 1 - A

Roll No: _____

Name: _____

Instructions: Do not write on the question paper or tick any option. Write selected option inside the brackets given at the end of each question "Answer []"

1. According to Chapter 1, what does the "P" in the PCS framework stand for?

A) Probability B) Predictability C) Processing D) Precision Answer []

2. In the context of Veridical Data Science, the transition from the "Real-World Realm" to the "Data Realm" relies primarily on:

A) Machine learning algorithms B) Data visualization techniques C) Measurement and data collection processes D) Computational hardware Answer []

3. According to the Data Science Life Cycle (DSLCL) in Chapter 2, what is typically the first step in a data science project?

A) Exploratory Data Analysis B) Data Cleaning C) Question Formulation D) Modeling Answer []

4. During the DSLCL, if a data scientist discovers that the available data cannot answer the original research question, what is the most appropriate next step?

A) Use a powerful machine learning model to find an answer. B) Refine the question or acquire new, appropriate data. C) Skip directly to Explanatory Data Analysis. D) Delete the missing values and proceed. Answer []

5. As discussed in Chapter 4, what is the defining characteristic of "Tidy Data"?

A) All missing values have been replaced with zeros. B) The data has been normalized to a standard scale. C) Every variable is a column, and every observation is a row. D) The dataset contains only numerical variables. Answer []

6. Why is it considered a best practice in data cleaning to encapsulate your cleaning steps into functions or scripts?

A) To make the data file size smaller. B) To ensure the process is reproducible and changes can be tracked. C) To permanently overwrite the original raw data. D) To prevent other researchers from seeing the raw data. Answer []

7. Based on Chapter 5, what is the primary goal of Exploratory Data Analysis (EDA)?

A) To present a final, polished conclusion to stakeholders. B) To discover underlying patterns, spot anomalies, and formulate hypotheses. C) To train predictive machine learning models. D) To prove that the initial hypothesis was correct. Answer []

8. According to Chapter 5, what is the recommended approach for starting an Exploratory Data Analysis (EDA)?

A) Applying advanced clustering algorithms before creating any visual plots. B) Removing all missing values and outliers before formulating any questions. C) Plotting every single variable against every other variable in the dataset. D) Following a question-and-answer workflow guided by domain knowledge. Answer []

9. Which of the following best describes the principle of "Stability" in the PCS framework (Chapter 1)?

A) The model must predict exactly the same outcome every single time, regardless of input. B) The data science pipeline should run without crashing. C) The conclusions drawn should remain robust even if human judgment calls or data samples change slightly. D) The data must never be altered once it is collected. Answer []

10. When a data scientist examines the summary statistics of a variable and visualizes its distribution to check for errors, they are bridging which two phases of the DSLC?

A) Question Formulation and Data Acquisition B) Data Cleaning and Exploratory Data Analysis C) Modeling and Reporting D) Data Acquisition and Modeling Answer []

Quiz A (Section A) Answers:

1. B (Predictability)
2. C (Measurement and data collection processes)
3. C (Question Formulation)
4. B (Refine the question or acquire new, appropriate data)
5. C (Every variable is a column, and every observation is a row)
6. B (To ensure the process is reproducible and changes can be tracked)
7. B (To discover underlying patterns, spot anomalies, and formulate hypotheses)
8. D (Following a question-and-answer workflow guided by domain knowledge.)
9. C (The conclusions drawn should remain robust even if human judgment calls or data samples change slightly)
10. B (Data Cleaning and Exploratory Data Analysis)